

Use of cationic surfactant to remove solvent-based ink from rigid high density polyethylene surfaces

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Abstract

Increasing space limitations for disposal of solid waste and shortages of raw materials for plastic have increased the need for recycling. However, printed plastic surfaces cannot be effectively recycled unless the inks are removed. The use of surfactants for deinking rather than organic solvents or chlorine bleaches is environmentally attractive. In this study, the effect of cationic surfactant concentration, pH, process temperature, pre-soaking time, shaking time and abrasion were investigated for blue solvent-based ink on rigid high density polyethylene surfaces. The cationic surfactant (*n*-hexadecyl trimethyl ammonium bromide) was effective only at concentrations above the critical micelle concentration (CMC) and at high pH levels (11.75 and above). Both below and above the CMC, ink removal was promoted by increasing temperature, pre-soaking time and shaking time. The presence of an abrasive also helped to detach the loosened ink from the plastic surface. Solubilization appears to be an important mechanism aiding deinking. © 2002 Elsevier Science B.V. All rights reserved.

Keywords: Surfactant; Deinking; Plastic container; Recycling; Solvent-based ink

1. Introduction

The use of rigid plastic packaging (e.g. milk bottles) has rapidly increased for many applications because of its light weight, strength, durability, moldability, colorability and relatively

low energy requirement in converting resin to manufactured products. In the future, increasing scarcity of raw materials for virgin plastic will make recycling more attractive. Moreover, most of the used plastics go to landfills, thereby increasing solid waste disposal problems. Plastic waste generation increases annually, but only a small proportion of this waste is recycled, forcing both governments and industries to seriously consider plastic recycling.

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At present, polymers produced from recycling are not as valuable as virgin polymers because the residual ink colors and changes the physical properties of the polymer [1]. These problems can be avoided if ink is removed from the plastic surface before recycling. Possible alternatives for plastic deinking include using either organic solvents or surfactants; however, the use of surfactants has an important advantage in being environmentally friendly.

The use of surfactants to remove ink (deink) from a polymer surface is an example of a surfactant-based separation process [2,3]. Deinking is fundamentally a laundering process. The adhesion of ink to plastic packaging materials and soil to clothing is due to van der Waals, electrostatic and chemical bonding, among other forces. Similar to laundering, deinking is a two step process: (i) detachment of soil (ink) from the substrate (plastic surface); and (ii) dispersion of soil in a washing bath by mechanical action [4,5]. The effect of surface tension, surfactant adsorption, wetting, dispersion and solubilization are all important in the deinking process. It is well known that single component liquids wet substrates that have higher critical surface tension values than the liquid surface tension [6–8]. During printing, the lower surface tension of the ink causes it to wet the polymer surface and adhere to the surface better [9]. The removal of ink from polymer surfaces with the use of aqueous solutions of surfactant is aided by adsorption of surfactants onto the polymer and ink surfaces, thereby decreasing the interfacial tension values of both polymer–water and ink–water. If the reduction of interfacial tension is so strong that the sum of the interfacial tensions of polymer–water and ink–water is equal to or less than the ink–polymer interfacial tension, it is thermodynamically favorable for the ink to detach from the polymer surface [5,10]. Rubbing of the polymer surface against another solid (e.g. other polymer or abrasive) and shearing forces from agitation help remove loosened ink and keep the detached ink dispersed [1,4,10–13]. Solubilization [14] of molecularly dispersed dyes or binder molecules into micelles can also contribute to deinking. In previous studies of deink-

ing of polyethylene film, solvent-based inks were found to be much more difficult to remove than water-based inks [1,11]. Studies of the effect of surfactant type on deinking of rigid polyethylene surfaces [12] and polyethylene film [1] have shown that cationic surfactants are more effective than anionic and nonionic surfactants and have equal or greater effectiveness as amphoteric surfactants. In this research, the effect of processing parameters on the effectiveness of deinking focused on the removal of screen-type solvent-based blue ink printed on flame-treated high density polyethylene (HDPE) blow molded bottles using a cationic surfactant.

2. Experimental

2.1. Materials

Plastic containers used for this research were all composed of HDPE and printed with blue color solvent-based ink by the screen printing process. Prior to printing, the containers were flame-treated in order for the ink to adhere well to the surface. All the containers were treated and printed by SVB Drinking Water Co., Thailand. Plastic samples ($\approx 8 \times 40$ mm) containing blue-colored printed ink were cut from the printed area of the plastic bottles. Small irregular shapes of ceramic pieces were used as the abrasive material in order to facilitate the detachment of loosened ink from the rigid plastic surfaces. The abrasive was composed of refractory insulators (used for electrical wire) of tubular shape ≈ 5 mm diameter and 5 mm in length. The hole in the center was ≈ 3 mm in diameter. Ceramic beads [1] or zirconium silicate grinding beads [13] have been used as abrasives in other studies.

The cationic surfactant, *n*-hexadecyl trimethyl ammonium bromide or cetyl trimethyl ammonium bromide (CTAB), having 98% purity in powder form, was supplied by Fluka Co., Switzerland. It was used as received without further purification. Sodium hydroxide with 98% purity, which was used for pH adjustment, was supplied by EKA Nobel Co., Sweden.

2.2. Analysis before deinking process

The amount of ink present on each sample before deinking was determined by using an optical scanning method. For optical scanning, each plastic sample was carefully positioned on the scanner and scanned at optimum conditions (standardized) using a HP scanJet 4c scanner. In order to prevent the reflection from the white surface of the scanner cover, a black poster board was placed behind the sample during scanning. After scanning, an image file was imported into an Adobe PhotoShop program to quantify the amount of ink (pixels) present on the plastic surface.

2.3. Deinking

Each CTAB solution was prepared with distilled water at room temperature and stirred with a magnetic stirrer to produce a homogeneous solution. The surfactant solution was warmed up slightly to 40 °C to speed up dissolution (since dissolution is slow in water at room temperature). The pH level of the surfactant solution was then adjusted by adding a sodium hydroxide solution from a burette. All pH measurements were carried out using a Benchtop pH/ISE meter Model 920A with Triode pH electrode Model 9157BN. After analyzing the samples (before deinking) by the optical scanning method, each sample was placed in a 15-ml surfactant solution in a conical flask. The samples were then allowed to soak for either 2 h in the presence of 25 irregular-shaped ceramic pieces (abrasive material) or for various times in the absence of ceramic pieces. All samples were allowed to soak by standing the flasks in a temperature controlled water bath. After soaking, the flasks were shaken in the temperature controlled shaking water bath (GFL model 1086) at 200 oscillations per minute. After shaking, the samples were removed from the flasks, washed several times with distilled water and air-dried. Each experiment was repeated at least twice.

2.4. Analysis after deinking process

The extent of deinking of the samples were first

evaluated visually. The deinked plastic samples were rescanned on the HP ScanJet scanner using the identical procedure and conditions as used for the original samples. The amount of ink removed (%) was determined using the following relationship:

Ink removed (%)

$$= \frac{[(\text{pixels}_{\text{before deinking}} - \text{pixels}_{\text{after deinking}})]}{\text{pixels}_{\text{before deinking}}} \times 100.$$

3. Results and discussion

3.1. Results

Base conditions were a pH of 12, a temperature of 30 °C, 2 h pre-soaking time, 2 h shaking time and absence of abrasive. Figs. 1–10 show photographs of deinked samples and ink removal for

CTAB concentration	Without abrasive	With abrasive
0	มพันธ์ทิพย์ มพันธ์ทิพย์	มพันธ์ทิพย์ มพันธ์ทิพย์
25% CMC	มพันธ์ทิพย์ มพันธ์ทิพย์	มพันธ์ทิพย์ มพันธ์ทิพย์
50% CMC	มพันธ์ทิพย์ มพันธ์ทิพย์	มพันธ์ทิพย์ มพันธ์ทิพย์
75% CMC	มพันธ์ทิพย์ มพันธ์ทิพย์	มพันธ์ทิพย์ มพันธ์ทิพย์
CMC	มพันธ์ทิพย์ มพันธ์ทิพย์	มพันธ์ทิพย์ มพันธ์ทิพย์
3 x CMC	มพันธ์ทิพย์ มพันธ์ทิพย์	มพันธ์ทิพย์ มพันธ์ทิพย์
5 x CMC		
7 x CMC		
10 x CMC		

Fig. 1. Deinked plastic surfaces that have been treated with and without abrasive material as a function of CTAB concentration.

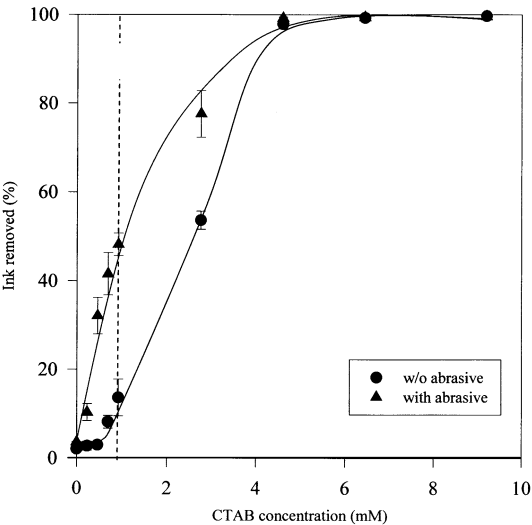


Fig. 2. Amount of ink removed from printed plastic surfaces with and without abrasive material as a function of CTAB concentration.

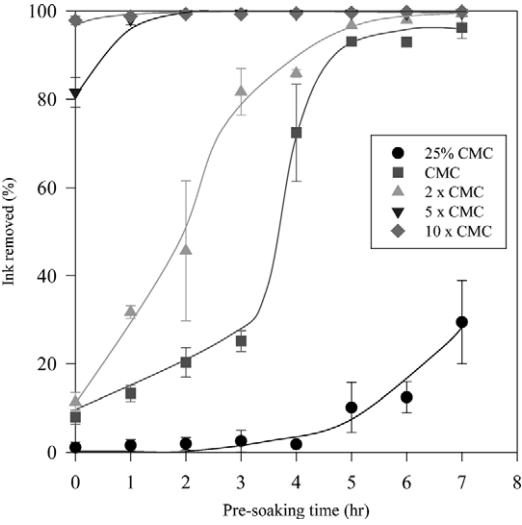


Fig. 4. Amount of ink removed from printed plastic surfaces at selected CTAB concentrations as a function of pre-soaking time.

the conditions studied. The critical micelle concentration (CMC) of CTAB in the absence of added electrolyte at 25 °C is 0.92 mM [15].

3.1.1. Effect of CTAB concentration and abrasive
As shown in Figs. 1 and 2, in the absence of ceramic pieces, there is no significant ink removal

Pre-soaking Time (hr)	25% CMC	CMC	2 x CMC	5 x CMC	10 x CMC
0	มพินธ์ทิพย์ มพินธ์ทิพย์	มพินธ์ทิพย์ มพินธ์ทิพย์	มพินธ์ทิพย์ มพินธ์ทิพย์	มพินธ์ทิพย์ มพินธ์ทิพย์	มพินธ์ทิพย์ มพินธ์ทิพย์
1	มพินธ์ทิพย์ มพินธ์ทิพย์	มพินธ์ทิพย์ มพินธ์ทิพย์	มพินธ์ทิพย์ มพินธ์ทิพย์	มพินธ์ทิพย์ มพินธ์ทิพย์	มพินธ์ทิพย์ มพินธ์ทิพย์
2	มพินธ์ทิพย์ มพินธ์ทิพย์	มพินธ์ทิพย์ มพินธ์ทิพย์	มพินธ์ทิพย์ มพินธ์ทิพย์	มพินธ์ทิพย์ มพินธ์ทิพย์	มพินธ์ทิพย์ มพินธ์ทิพย์
3	มพินธ์ทิพย์ มพินธ์ทิพย์	มพินธ์ทิพย์ มพินธ์ทิพย์	มพินธ์ทิพย์ มพินธ์ทิพย์	มพินธ์ทิพย์ มพินธ์ทิพย์	มพินธ์ทิพย์ มพินธ์ทิพย์
4	มพินธ์ทิพย์ มพินธ์ทิพย์	มพินธ์ทิพย์ มพินธ์ทิพย์	มพินธ์ทิพย์ มพินธ์ทิพย์	มพินธ์ทิพย์ มพินธ์ทิพย์	มพินธ์ทิพย์ มพินธ์ทิพย์
5	มพินธ์ทิพย์ มพินธ์ทิพย์	มพินธ์ทิพย์ มพินธ์ทิพย์	มพินธ์ทิพย์ มพินธ์ทิพย์	มพินธ์ทิพย์ มพินธ์ทิพย์	มพินธ์ทิพย์ มพินธ์ทิพย์
6	มพินธ์ทิพย์ มพินธ์ทิพย์	มพินธ์ทิพย์ มพินธ์ทิพย์	มพินธ์ทิพย์ มพินธ์ทิพย์	มพินธ์ทิพย์ มพินธ์ทิพย์	มพินธ์ทิพย์ มพินธ์ทิพย์
7	มพินธ์ทิพย์ มพินธ์ทิพย์	มพินธ์ทิพย์ มพินธ์ทิพย์	มพินธ์ทิพย์ มพินธ์ทิพย์	มพินธ์ทิพย์ มพินธ์ทิพย์	มพินธ์ทิพย์ มพินธ์ทิพย์

Fig. 3. Samples of deinked plastic surfaces treated with selected CTAB concentrations as a function of pre-soaking time.

from the plastic surfaces below the CMC, while there is substantial removal when abrasives are present, except when no surfactant is present and very little removal occurs. There is a CTAB concentration range in which deinking increases rapidly to 100%; in the absence of abrasives, this concentration range is $\approx 150\%$ higher than in the presence of abrasives. For example, 50% deinking occurs at 1.0 mM CTAB with abrasives and at 2.5 mM CTAB without abrasives. In either case, complete (100%) deinking occurs at a CTAB concentration of 5 mM. So, abrasives speed up deinking, but are not necessary for complete deinking. The separation of abrasives from recovered plastic and from ink might make abrasives undesirable. If abrasives are discarded with detached ink, this results in an increase in the mass of solid waste requiring disposal.

3.1.2. Effect of pre-soaking time

As shown in Figs. 3 and 4, the shaking time required to attain a specified degree of deinking can be substantially reduced by pre-soaking. The effect is particularly pronounced at CTAB concentrations around the CMC. For example, at CTAB concentrations of $2 \times \text{CMC}$, ink removal increases from 10 to 80% as pre-soaking time increases from 0 to 3 h. Deinking efficiency can be dramatically affected at specific pre-soaking times in some cases. For example, for a pre-soaking time increase from 3 to 5 h at the CMC, ink removal increased from 25 to 90%.

3.1.3. Effect of shaking time

Figs. 5 and 6 show that shaking time can have a substantial effect on ink removal. Comparing Figs. 4 and 6, there is sometimes an inhibition

Shaking Time (hr)	25% CMC	CMC	5 x CMC	10 x CMC
0	มพันธ์ทิพย์ มพันธ์ทิพย์	มพันธ์ทิพย์ มพันธ์ทิพย์	มพันธ์ทิพย์ มพันธ์ทิพย์	มพันธ์ทิพย์ มพันธ์ทิพย์
1	มพันธ์ทิพย์ มพันธ์ทิพย์	มพันธ์ทิพย์ มพันธ์ทิพย์	มพันธ์ทิพย์ มพันธ์ทิพย์	
2	มพันธ์ทิพย์ มพันธ์ทิพย์	มพันธ์ทิพย์ มพันธ์ทิพย์		
3	มพันธ์ทิพย์ มพันธ์ทิพย์	มพันธ์ทิพย์ มพันธ์ทิพย์		
4	มพันธ์ทิพย์ มพันธ์ทิพย์	มพันธ์ทิพย์ มพันธ์ทิพย์		
5	มพันธ์ทิพย์ มพันธ์ทิพย์			
6	มพันธ์ทิพย์ มพันธ์ทิพย์			
7	มพันธ์ทิพย์ มพันธ์ทิพย์			

Fig. 5. Samples of deinked plastic surfaces treated with selected CTAB concentrations as a function of shaking time.

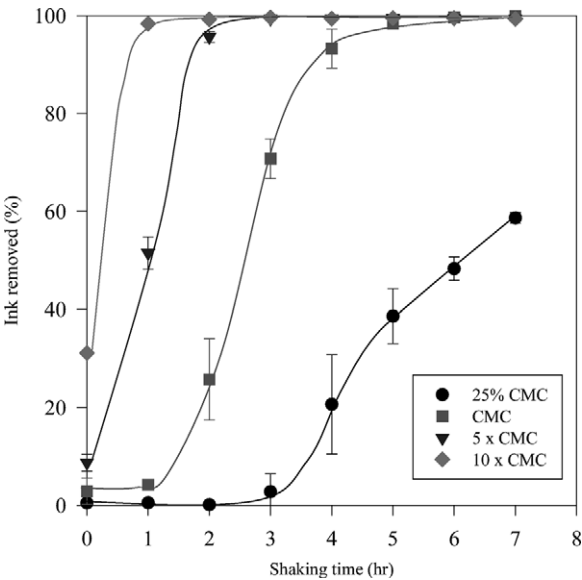


Fig. 6. Amount of ink removed from printed plastic surfaces at selected CTAB concentrations as a function of shaking time.

period of time for both pre-soaking and shaking, during which little deinking occurs, followed by a narrow period of time (we will call this the detachment period) during which ink removal increases rapidly to near completeness. From a design viewpoint, it is important to choose residence times for pre-soaking and shaking which are at the end of the detachment periods for each. Longer pre-soaking times permit shorter shaking

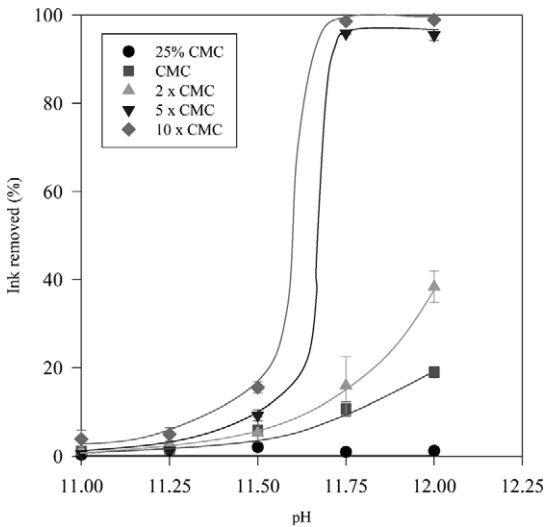


Fig. 8. Amount of ink removed from printed plastic surfaces at selected CTAB concentrations as a function of pH.

times and longer shaking times permit shorter pre-soaking times, so there is an economic optimization required to determine design conditions. For example, > 90% ink removal can be attained at a CTAB concentration at the CMC with 5 h pre-soaking and a standard 2 h shaking time, while with 2 h pre-soaking, 4 h shaking time is needed to yield > 90% deinking. Since pre-soaking tanks generally require much less capital expense than agitated process units, pre-soaking can improve process economics.

pH	25% CMC	CMC	2 x CMC	5 x CMC	10 x CMC
11.00	มพินธ์ทิพย์ มพินธ์ทิพย์	มพินธ์ทิพย์ มพินธ์ทิพย์	มพินธ์ทิพย์ มพินธ์ทิพย์	มพินธ์ทิพย์ มพินธ์ทิพย์	มพินธ์ทิพย์ มพินธ์ทิพย์
11.25	มพินธ์ทิพย์ มพินธ์ทิพย์	มพินธ์ทิพย์ มพินธ์ทิพย์	มพินธ์ทิพย์ มพินธ์ทิพย์	มพินธ์ทิพย์ มพินธ์ทิพย์	มพินธ์ทิพย์ มพินธ์ทิพย์
11.50	มพินธ์ทิพย์ มพินธ์ทิพย์	มพินธ์ทิพย์ มพินธ์ทิพย์	มพินธ์ทิพย์ มพินธ์ทิพย์	มพินธ์ทิพย์ มพินธ์ทิพย์	มพินธ์ทิพย์ มพินธ์ทิพย์
11.75	มพินธ์ทิพย์ มพินธ์ทิพย์	มพินธ์ทิพย์ มพินธ์ทิพย์	มพินธ์ทิพย์ มพินธ์ทิพย์	มพินธ์ทิพย์ มพินธ์ทิพย์	มพินธ์ทิพย์ มพินธ์ทิพย์
12.00	มพินธ์ทิพย์ มพินธ์ทิพย์	มพินธ์ทิพย์ มพินธ์ทิพย์	มพินธ์ทิพย์ มพินธ์ทิพย์	มพินธ์ทิพย์ มพินธ์ทิพย์	มพินธ์ทิพย์ มพินธ์ทิพย์

Fig. 7. Samples of deinked plastic surfaces treated with selected CTAB concentrations as a function of pH.

T (°C)	25% CMC	CMC	2 x CMC	5 x CMC	10 x CMC
30	มพณธ์ทพย มพณธ์ทพย	มพณธ์ทพย มพณธ์ทพย	มพณธ์ทพย มพณธ์ทพย		
35	มพณธ์ทพย มพณธ์ทพย	มพณธ์ทพย มพณธ์ทพย	มพณธ์ทพย มพณธ์ทพย		
40	มพณธ์ทพย มพณธ์ทพย	มพณธ์ทพย มพณธ์ทพย	มพณธ์ทพย มพณธ์ทพย		
45	มพณธ์ทพย มพณธ์ทพย	มพณธ์ทพย มพณธ์ทพย	มพณธ์ทพย มพณธ์ทพย		

Fig. 9. Samples of deinked plastic surfaces treated with selected CTAB concentrations as a function of temperature.

3.1.4. Effect of pH

The effect of pH on ink removal is shown in Figs. 7 and 8, where a dramatic effect of pH is observed. At a pH of 11, almost no deinking is observed at any CTAB concentration. Yet, at a CTAB concentration of $5 \times \text{CMC}$ and above, at a pH of 11.75 and above, almost complete deinking is observed. High pH (at least 11.5) is crucial for effective ink removal. Of course, under these extremely basic conditions, materials constraints increase process equipment costs substantially. As will be discussed for temperature, pH effects on CMC were not accounted for and are assumed to be small at the added electrolyte levels ($\leq 0.01 \text{ M}$) from added NaOH.

3.1.5. Effect of temperature

Higher temperatures can substantially increase deinking efficiency, as shown in Figs. 9 and 10. At 45°C , $>90\%$ ink removal was observed even at a CTAB concentration of only 25% of the CMC. Higher temperatures can increase energy costs and require robustness of process equipment materials (particularly at the high pH levels anticipated). Note that the CMC referred to here is that at 25°C . Since the CMC is only mildly dependent on temperature for ionic surfactants [15], temperature corrections were not made. So, for example, when the CTAB concentration is referred to as at the CMC, this is not entirely accurate at 45°C . In terms of the fairly large concentration increments used here, this is not a significant consideration.

Qualitative observations give some insight into the mechanisms of deinking. When significant deinking occurred, the wash solution was always blue in color. The greater the degree of deinking, the bluer the color. At 30°C , particles were observed in solution and the remaining ink on the polymer surface was hard. At 45°C , the particles in the solution were observed to be larger than at 30°C . At 45°C , the residual ink on the polymer was soft and sticky.

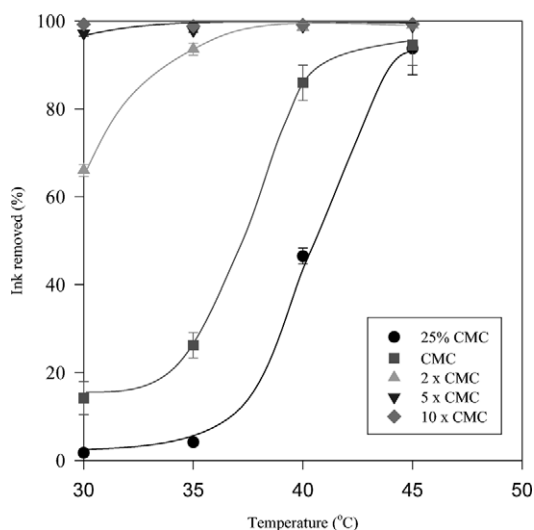


Fig. 10. Amount of ink removed from printed plastic surfaces at selected CTAB concentrations as a function of temperature.

3.2. Mechanisms of deinking

One problem with understanding physical mechanisms of ink removal is the ill-defined nature of the inks and binder due to their complexity and due to the proprietary nature of their formulation. However, the results here can give some insight.

The surfactant can aid detachment by mechanisms such as adsorption on ink and plastic surfaces, decreasing ink/water and plastic/water interfacial tensions, making detachment more thermodynamically favorable. This is a common mechanism aiding detergency of both particulate and oily soils [5,16]. Surfactant can sometimes 'liquefy' soil, an ill-defined term implying that the soil can either flow and be removed by the roll-back mechanism [4,5,10,16] or can be shaken up into little pieces as surfactant diffuses or penetrates into the soil and loosens it up. The surfactant can also improve wetting, aiding a water layer to form between ink and polymer surface. The wetting enhancement is due to surfactant adsorption on ink and plastic-trapped air is displaced. The surfactant can aid in anti-redeposition (preventing the ink from readSORbing) by dispersing solid ink particles by adsorbing in the ink surface, emulsifying liquid ink droplets and solubilizing ink molecules in micelles [1,5,11,14,17].

The surface area of the polymer and ink are low enough that the final concentration of surfactant in solution is probably not significantly lower than the initial concentration. Above the CMC, the primary function of incremental surfactant is to solubilize ink molecules within micelles. A secondary advantage could be the micelles disassociating to provide surfactant monomer, speeding up rate-driven processes. However, over the time periods covered here (hours), this is generally not a significant factor, so solubilization can be deduced as an important mechanism of ink uptake by the solution. This implies that a significant amount of ink is molecularly dispersed in order to be solubilized, instead of being present as particles or droplets. Another possibility is that the micelles are solubilizing the ink binder.

Both pre-soaking and shaking time are important. It is possible that either the rate of surfactant adsorption or rate of penetration or liquefaction can be the rate-determining step here. The beneficial effect of abrasives at short times imply that it is the rate of detachment of ink after surfactant adsorption has occurred which may be limiting, also consistent with no significant deinking with abrasive, but no surfactant. Solubilization is a very fast process [18] and is probably not rate-determining.

Detergency is generally more effective at high pH, as more negative charges are imposed on the contaminant and on the substrate, making them more repulsive from each other [5,8,10]. This would make ink detachment and antiredeposition easier in this case. The ink particles can be attached to the surface by binders. Attenuated total reflection (ATR) FTIR [12] indicated that the ink binder contained ester groups for this system. The intense C=O stretching bond of saturated aliphatic ester groups is in the frequency range of 1750–1735 cm^{-1} [19] in agreement with the spectrum observed for this ink. Sodium hydroxide, which was used for pH adjustment, is hypothesized to break up the ink binder by hydrolyzing the ester groups, especially at pH 12 [11,12]. A study of selective froth flotation to separate two polymers from each other has shown that the polymer surface hydrophobicity can be dependent on pH [20]. This wettability dependence on pH may be a factor here and will be a focus of future work.

The improvement in deinking with increasing temperature is probably due to rate effects, since surfactant adsorption decreases with increasing temperature [21] and solubilization is only mildly temperature dependent [22].

4. Conclusions

Increasing pH, temperature, pre-soaking time, shaking time, surfactant concentration and adding abrasives improve deinking. Complete deinking is attainable under reasonable process conditions and this work shows some variable effects useful in process optimization.

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